Examiner only

2. Caesium has many isotopes.
One of these is caesium-137.

(a) The symbol for the isotope caesium-137 is $^{137}_{55}Cs$.

Tick (✓) the box next to the symbol of another isotope of caesium. [1]

$^{137}_{54}Cs$ ☐

$^{133}_{55}Cs$ ☐

$^{138}_{57}Cs$ ☐

(b) When caesium-137 decays, beta particles are emitted from unstable nuclei.

(i) Tick (✓) the box next to the correct statement about unstable nuclei. [1]

They have too many electrons ☐

They have too many protons ☐

The number of neutrons and protons is unbalanced ☐

(ii) Tick (✓) the box next to the correct symbol for a beta particle. [1]

$^{1}_{-1}\beta$ ☐

$^{0}_{-1}\beta$ ☐

$^{-1}_{0}\beta$ ☐

(iii) Tick (✓) the box next to the correct statement about a beta particle. [1]

It is a helium nucleus ☐

It is an electromagnetic wave ☐

It is a high energy electron ☐

Examiner only

(c) The activity of a sample of caesium-137 is 160 units.
It has a half-life of 30 years.

(i) Underline **one** number in the brackets to complete the sentence. [1]

After 30 years, the activity of the caesium-137 sample will be (**20** / **40** / **80**) units.

(ii) Jason says the activity of the sample will be zero after 60 years.
Explain whether you agree with Jason. [2]

....

....

....

7

3420U201
05